MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, FLAME RETARDED, MODIFIED, POLYOLEFIN, FLEXIBLE, CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

REQUIREMENTS:

Continuous operating temperature range: -55°C (-67°F) to +120°C (+250°F).

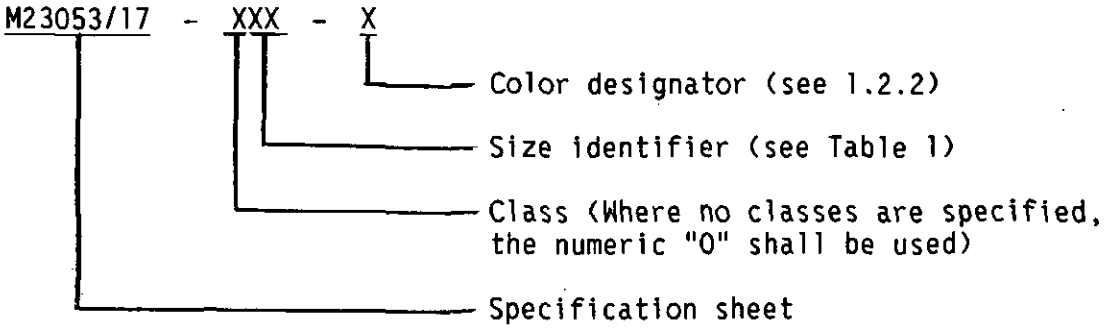
Classification: The heat shrinkable sleeving shall be furnished in the following classes, as specified (see 6.2.1):

Class 1 - Standard wall
Class 2 - Thin wall

Color: Translucent to clear.

Longitudinal Change: Classes 1 and 2: +2, -10 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



Example: Class 1, 0.250 inch (6.35 mm) as supplied ID sleeving shall be identified as M23053/17-106-C.

NOTE: Entire specification sheet revised.

AMSC N/A FSC 5970

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TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/17-101-C	.046(1.17)	.023(.58)	.016 ± .003(.406 ± .076)
M23053/17-102-C	.063(1.60)	.031(.79)	.017 ± .003(.432 ± .076)
M23053/17-103-C	.093(2.36)	.046(1.17)	.020 ± .003(.508 ± .076)
M23053/17-104-C	.125(3.18)	.062(1.58)	.020 ± .003(.508 ± .076)
M23053/17-105-C	.187(4.75)	.093(2.36)	.020 ± .003(.508 ± .076)
M23053/17-106-C	.250(6.35)	.125(3.18)	.025 ± .003(.635 ± .076)
M23053/17-107-C	.375(9.53)	.187(4.75)	.025 ± .003(.635 ± .076)
M23053/17-108-C	.500(12.7)	.250(6.35)	.025 ± .003(.635 ± .076)
M23053/17-109-C	.750(19.1)	.375(9.53)	.030 ± .003(.762 ± .076)
M23053/17-110-C	1.000(25.4)	.500(12.7)	.035 ± .005(.889 ± .127)
Class 2			
M23053/17-201-C	.046(1.17)	.023(.58)	.010 ± .003(.254 ± .076)
M23053/17-202-C	.063(1.60)	.031(.79)	.010 ± .003(.254 ± .076)
M23053/17-203-C	.093(2.36)	.046(1.17)	.012 ± .003(.305 ± .076)
M23053/17-204-C	.125(3.18)	.062(1.58)	.012 ± .003(.305 ± .076)
M23053/17-205-C	.187(4.75)	.093(2.36)	.012 ± .003(.305 ± .076)
M23053/17-206-C	.250(6.35)	.125(3.18)	.014 ± .003(.356 ± .076)
M23053/17-207-C	.375(9.53)	.187(4.75)	.014 ± .003(.356 ± .076)
M23053/17-208-C	.500(12.7)	.250(6.35)	.014 ± .003(.356 ± .076)
M23053/17-209-C	.750(19.1)	.375(9.53)	.017 ± .003(.432 ± .076)
M23053/17-210-C	1.000(25.4)	.500(12.7)	.020 ± .005(.508 ± .127)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

Unrestricted shrinkage: Test method 4.6.5: 200° ± 2°C (392° ± 4°F) for 3 minutes.

TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, *minimum*	Table 1	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 200° ± 3°C (392° ± 6°F)

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Secant modulus, psi (MPa), minimum	15,000(103.4)	4.6.13.1, 2 percent strain, ASTM D 882
Restricted shrinkage	No cracks	4.6.6.1.1 200° ± 2°C (392° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Clarity stability	Pass	4.6.17 125° ± 1°C (257° ± 2°F), 24 hours
Concentricity, min.	70%	4.6.3.3
After unrestricted shrinkage:		
ID, maximum	Table 1	4.6.3
Wall thickness	Table 1	4.6.3
Tensile strength, psi (MPa), minimum	1,500(10.4)	4.6.14 ASTM D 638, 20 inches/min.
Ultimate elongation, percent, minimum	200	4.6.14 ASTM D 638, 20 inches/min.
Dielectric strength, volts/mil (Kv/mm), minimum	750(29.5)	4.6.2 ASTM D 2671
Volume resistivity, ohm-CM, minimum	10^{14}	4.6.2 ASTM D 876
Dielectric constant, maximum	3.7	4.6.2 ASTM D 150
Specific gravity, maximum	1.35	4.6.2 ASTM D 792
Water absorption, 23°C percent, maximum	0.5	4.6.2 ASTM D 570, 24 hrs. at
Corrosion	No corrosion	4.6.10.1 and 4.6.10.2 120° ± 2°C (248° ± 4°F), 16 hours
Low temperature flexibility	No cracking	4.6.7.1 -55° ± 1°C (-67° ± 2°F)
Flammability	Self-extinguishing within 15 seconds; 3 inches	4.6.15.2

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Heat resistance, property after:		4.6.9 150° ± 2°C (302° ± 4°F), 168 hours
Ultimate elongation, percent, minimum	100	
Fluid resistance, property after:		4.6.11
Tensile strength, psi (MPa), minimum	1000(6.9)	
Elongation, percent, minimum	100	
Dielectric strength, volt/mil (Kv/mm), minimum	500(12.7)	
Fungus resistance	Non-nutrient	4.6.2 ASTM G 21

1/ Unless otherwise specified, requirements, test procedures, and conditions are for all classes.

Shelf life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for 2 years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use:

Class 1 sleevings are intended for normal use where flame retarded translucent to clear materials are required.

Class 2 is intended where thin wall product still provides for adequate electrical insulation and confined spaces require smaller wire diameters.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

(Project 5970-0577-16)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-23053 /17A	INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABL GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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